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Digital systems and methods for high precision control in nuclear reactors

Abstract

Control rod drives include all-digital monitoring, powering, and controlling systems for operating the drives. Each controlling system includes distinct microprocessor-driven channels that independently monitor and handle control rod drive position information reported from multiple position sensors per drive. Controlling systems function as rod control and information systems with top-level hardware interfaced with nuclear plant operators other plant systems. The top-level hardware can receive operator instructions and report control rod position, as well as report errors detected using redundant data from the multiple sensors. Positional data received from each drive is multiplexed across plural, redundant channels to allow verification of the system using independent position data as well as operation of the system should a single channel or detector fail. Control rod drives are capable of positioning and detecting position of control elements in fine increments, such as 3-millimeter increments, with plural position sensors that digitally report drive status and position.

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation, and claims the benefit under 35 U.S.C. § 120, of U.S. application Ser. No. 17/131,200, filed Dec. 22, 2020, which is a divisional, and claims the benefit under 35 U.S.C. § § 120 & 121, of U.S. application Ser. No. 15/453,195 filed Mar. 8, 2017, now U.S. Pat. No. 10,910,115. All these applications are incorporated by reference herein in their entireties.

BACKGROUND
(1) FIG. **1** is a cross-sectional illustration of a related art control rod drive **10** useable in a nuclear reactor. For example, related art control rod drive **10** may be a Fine Motion Control Rod Drive (FMCRD) as used in next generation BWRs like ESBWRs. As shown in FIG. **1**, control rod drive is housed outside, such as below, reactor pressure vessel **1**, where drive **10** inserts or withdraws a control element, such as a control blade (not shown), attached thereto via bayonet coupling **15**. Hollow piston **16** connects to bayonet coupling **15** and is vertically moveable inside of an outer tube in drive **10**. Drive **10** includes ball screw **11** coupled with ball nut **12** that drives screw **11** vertically when rotated, such that hollow piston **16**, bayonet coupling **15**, and the control element attached thereto may be positioned with precision at desired positions in a nuclear reactor core. During scram, hollow piston **16** may be lifted off ball nut **12** by hydraulic pressure in the outer tube, permitting rapid movement and insertion of the control element. A magnetic coupling **17** pairs internal and external magnets to rotate ball screw **11** across a pressure barrier, and a motor and brake **19** mounted below drive and stop magnetic coupling **17** to desired positioning.
(2) Power and control is provided to control rod drive by a Rod Control & Information System (RC&IS) that in varying conventional designs uses a mix of analog systems and digital transducers, including resolvers and synchros and encoders, and any required analogue-to-digital converters, coupled to motor and brake **19** to provide position detection, control, and power to the same. Several existing mixed analog and digital systems are able to detect and resolve position of associated control elements to several centimeters, with coarser position control. Related descriptions of drive **10** and FMCRD technology are found in GE-Hitachi Nuclear Energy, “The ESBWR Plant General Description,” Chapter 3—Nuclear Steam Supply Systems, Control Rod Drive System, Jun. 1, 2011, incorporated herein by reference in its entirety.
SUMMARY
(3) Example embodiments include all-digital control rod drives and associated systems for monitoring, powering, and controlling the drives. Control and information systems may be divided into distinct channels with switches and controls of each channel performing independent and redundant control rod drive monitoring and controlling. Control and information systems may separately be divided among main control logic associated with plant operators and top-level input

from other plant systems, remote cabinet equipment including multiplexed data handling for transmission to the main control logic, fine motion controllers associated with each control rod drive, and the control rod drive themselves. Each of these subsystems may pass control rod drive position information from multiple position sensors to the plant operators and other plant systems. Each piece of position information from an individual sensor, such as a digital position transducer, may be handled by one of the distinct channels of the control and information systems. Because the entire system, including control rod drives and control and information systems, may work with digital information on computer hardware processors and associated memories and busses, no digital-to-analog or analog-to-digital converter is necessary. By connecting and multiplexing each channel throughout the system, controllers associated with each channel may verify accuracy of position information with controllers of other channels, and redundant commands and data may be transmitted and received even if one channel or sensor fails.

(4) Control rod drives useable in example embodiments include fine motion control rod drives capable of positioning and detecting position of control elements in very fine increments, such as 3-millimeter increments, such as with rotating servo motors. Example control rod drives may further include several position sensors that digitally report control rod drive status and insertion position in these fine increments. A single control and information system may receive the multiple position sensor readings from multiple control drive systems and control the same using appropriate powering and control signals to achieve desired control rod positioning.

Description

BRIEF DESCRIPTIONS OF THE DRAWINGS

(1) Example embodiments will become more apparent by describing, in detail, the attached drawings, wherein like elements are represented by like reference numerals, which are given by way of illustration only and thus do not limit the terms which they depict.

(2) FIG. **1** is an illustration of a related art nuclear control rod drive.

(3) FIG. **2** is a schematic of a fine motion control rod drive motor and its motor control cabinet.

(4) FIG. **3** is a schematic of an example embodiment digital rod control and information system.

DETAILED DESCRIPTION

(5) Because this is a patent document, general broad rules of construction should be applied when reading it. Everything described and shown in this document is an example of subject matter falling within the scope of the claims, appended below. Any specific structural and functional details disclosed herein are merely for purposes of describing how to make and use examples. Several different embodiments and methods not specifically disclosed herein may fall within the claim scope; as such, the claims may be embodied in many alternate forms and should not be construed as limited to only examples set forth herein.

(6) It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited to any order by these terms. These terms are used only to distinguish one element from another; where there are “second” or higher ordinals, there merely must be that many number of elements, without necessarily any difference or other relationship. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of example embodiments or methods. As used herein, the term “and/or” includes all combinations of one or more of the associated listed items. The use of “etc.” is defined as “et cetera” and indicates the inclusion of all other elements belonging to the same group of the preceding items, in any “and/or” combination(s).

(7) It will be understood that when an element is referred to as being “connected,” “coupled,” “mated,” “attached,” “fixed,” etc. to another element, it can be directly connected to the other

element, or intervening elements may be present. In contrast, when an element is referred to as being "directly connected," "directly coupled," etc. to another element, there are no intervening elements present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., "between" versus "directly between," "adjacent" versus "directly adjacent," etc.). Similarly, a term such as "communicatively connected" includes all variations of information exchange and routing between two electronic devices, including intermediary devices, networks, etc., connected wirelessly or not.

(8) As used herein, the singular forms "a," "an," and "the" are intended to include both the singular and plural forms, unless the language explicitly indicates otherwise. It will be further understood that the terms "comprises," "comprising," "includes," and/or "including," when used herein, specify the presence of stated features, characteristics, steps, operations, elements, and/or components, but do not themselves preclude the presence or addition of one or more other features, characteristics, steps, operations, elements, components, and/or groups thereof.

(9) The structures and operations discussed below may occur out of the order described and/or noted in the figures. For example, two operations and/or figures shown in succession may in fact be executed concurrently or may sometimes be executed in the reverse order, depending upon the functionality/acts involved. Similarly, individual operations within example methods described below may be executed repetitively, individually or sequentially, so as to provide looping or other series of operations aside from single operations described below. It should be presumed that any embodiment or method having features and functionality described below, in any workable combination, falls within the scope of example embodiments.

(10) The inventors have recognized that existing locking collet piston and magnetic jack type control rod drives lack precision, being able to move in increments of 6 inches or more. Typical ESBWRs and other large, natural-circulation-dependent reactors require greater precision for reactivity control. Related FMCRD, such as in the ABWR, while having greater precision, are operated using analog sensors such as synchros and servos, which require an analog-to-digital converter for control. The use of analog sensors results in decrease of precision accuracy. Moreover, controls for FMCRD need to have very high reliability in nuclear applications, and single-channel analog controls represent an impermissible failure risk for lack of redundancy. Example embodiments described below address these and other problems recognized by the Inventors with unique solutions enabled by example embodiments.

(11) The present invention is a control rod drive control and information system. In contrast to the present invention, the small number of example embodiments and example methods discussed below illustrate just a subset of the variety of different configurations that can be used as and/or in connection with the present invention.

(12) Example embodiment control rod drive (CRD) control and information systems are useable with several different types of CRDs. Example embodiments are further useable with high-precision CRDs, including FMCRDs. Further, co-owned applications 62/361,628, filed Jul. 28, 2016 by Morgan et al., 62/361,625, filed Jul. 13, 2016 by Morgan et al., and 62/361,604, filed Jul. 13, 2016 by Morgan et al. are incorporated herein by reference in their entireties. These incorporated applications describe high-precision CRDs that are useable with example embodiments; for example, example embodiment control systems may be interfaced with, and control, the control rod drives disclosed in the incorporated applications.

(13) FIG. **2** is a schematic of FMCRD **10** where motor **19** and output of the FMCRD interface with an example embodiment system **100**. As shown in FIG. **2**, motor **19** may receive power and operative signals from a motor control cabinet **21** over motor interface **26**. For example, motor interface **26** may be radiation-hardened power cables that deliver electrical power, such as three-phase pulse-width modulated electrical signals, and/or information like speed controls to motor **19** in challenging environments such as operating nuclear reactor environments. Fine motion motor controller **20** may receive such power from a plant power source **25**, including main plant electrical

bus, local battery, emergency generators, or any other source of power.
(14) Motor controller **20** is configured to power motor interface **26** for both very fine intervals and very large power jumps. In this way, motor **19** may be powered to move a control blade attached to FMCRD **10** at very precise intervals, such as 3 millimeter vertical insertions or withdrawals, as well as rapid, large insertion strokes for shutdown. For example, motor **19** may be a servo motor that is controllable up to ¼.sup.th of one revolution of a ball screw in FMCRD **10** that can self-brake and provide long-term static torque. A full revolution may equate to 12 millimeters of vertical control element movement, so positioning on an order of a few millimeters is possible. Delivery of a precise amount of power through motor interface **26** may thus achieve desired precise motor actuation and control element positioning.
(15) FMCRD **10** may include multiple position detectors **18***a* and **18***b*, such as precise digital transducers that can determine an exact vertical position of ball nut **12** and/or other connective structures to report an accurate position of a control element within one-thirty-sixth of a millimeter. Position detectors **18***a* and **18***b* may be redundant or measure position of related structures as a verification of control element positioning. For example, two digital transducers as position detectors **18***a* and **18***b* may be coupled directly to a power output or shaft of motor **19** that encode rotations of the same to digitally represent motor **19** position and thus control element position. Because position detectors **18***a* and **18***b* may give digital output, informational positional signal interfaces **27***a* and **27***b* may be any connection capable of carrying digital signals, including fiber optic cable, coaxial cables, wireless signals, etc., separate or combined with motor interface **26**.
(16) Fine motion motor controller **20** includes two channel interfaces **28***a* and **28***b* each configured to receive digital position information from an associated one of detectors **18***a* and **18***b* via positional signal interfaces **27***a* and **27***b*. Channel interfaces **28***a* and **28***b* may separately handle individual positional signals and control inputs so as to preserve integrity of a redundant system and allow different signals to be verified against one another without intermixing or single failure affecting both signals. Fine motion motor control cabinet **21** housing motor controller **20** and various interfaces and communicative connections may be local or remote from FMCRD **10**. Because all components of cabinet **20** may be digital with no analog-to-digital converter, cabinet **20** may be relatively small and better hardened against an operating nuclear reactor environment.
(17) FIG. **3** is a schematic diagram of an example embodiment digital rod control and information system (RC&IS) **100**. As shown in FIG. **3**, example embodiment RC&IS **100** interfaces with a FMCRD **10** and fine motion motor controller **20** (FIG. **2**) through remote links **37***a* and **37***b*. RC&IS **100** may be located closer to an operator or control room, outside of containment and remote from FMCRD **10** and its motor control cabinet **21** (FIG. **2**), in which case remote links **37***a* and **37***b* may be any kind of digital data connection, including higher-reliability fiber optic cables or wireless connections. Remote links **37***a* and **37***b* together communicate dual-channel positional data from an FMCRD, such as digital positional data generated by position detectors **18***a* and **18***b* for redundant position determination of individual control elements.
(18) RC&IS **100** includes remote communication cabinet equipment **110** networked with an RC&IS main control logic **111** that is configured to receive input **200** from operators and other plant systems for control element operation and positioning. RC&IS **100** is completely digital, using microprocessors, field-programmable gate arrays, and digital communications for higher reliability and less space requirement. Cabinet equipment **110** may be all stored in a discreet and smaller cabinet remote or local to a control rod drive. For example, cabinet equipment **110**, lacking any analog-to-digital converter, may fit in a relatively small space and handle control element operations for a quarter of a nuclear core. Moreover, all-digital RC&IS **100** is less susceptible to noise and vibration found in an operating nuclear operating environment and thus can be located anywhere in a nuclear plant and can perform self-testing and alerting functions to maintain desired operations with lower failure rates than analog equipment.
(19) Example embodiment RC&IS **100** is configured to handle and control an FMCRD with dual,

redundant inputs to maintain operations in the case of single-channel failure, such as if a single position detector becomes unusable or signals are lost from one channel in example embodiments. As shown in FIG. **3**, RC&IS **100** includes two independent channels **120** and **130** in its cabinet equipment **110**, each in communication with an individual position data source such as remote links **37***a* and **37***b*. Although shown in communication with a single set of remote links **37***a* and **37***b*, it is understood that multiple FMCRDs and related motor controls may be input into and controlled by cabinet equipment **110**, including an entire core quadrant's worth of control drives.

(20) Each channel **120** and **130** independently handles distinct digital position and operation signals; however, channels **120** and **130** are also connected in order to verify and potentially replace one another. Bus communications **115** between all elements of channels **120** and **130** may be typical digital communications lines, including internal computer buses, motherboard connections, wired connections, or wireless connections. Each channel **120** and **130** includes multiplexed and redundant switches **121**, **122**, **124**, **131**, **132**, and **134**. Switches **122**, **124**, **132**, and **134** may receive positional data and issue motor command signals, as well as other data, for each control rod drive associated with RC&IS **100**. As shown by bus communications **115**, switches **121** and **122** associated with channel **120** may include connection to switches **131** and **132** of the other channel **130**. Thus, if a switch fails in either channel, or if other channel components become inoperative, the operative channel and switches therein may still receive, transmit, and otherwise handle all control information.

(21) Each channel **120** and **130** includes a controller **123** and controller **133**, respectively. The controllers **123** and **133** may receive, buffer, issue, and interpret control rod drive operational data. For example, controllers **123** and **133** may receive rod movement commands from RC&IS control channel controllers **165** and **155**, and such commands may be interpreted and sent to one or more motor controllers **20** (FIG. **2**) through remote links **37***a* and **37***b*. Motor controller **20** (FIG. **2**) may then provide power transmission to a motor **19** through motor interface **21** (FIG. **2**) to achieve desired driving, braking, and position from detectors **18***a* and **18***b* (FIG. **2**), interpreted for operator readout.

(22) As shown in FIG. **3**, controllers **123** and **133** may also perform comparison of rod movement commands from the RC&IS control logic controllers **155** and **165** to ensure that inconsistent rod movement commands do not result in rod movement. Switches **124** and **134** may provide redundant data links between each RC&IS logic channel controller **165** and **155**, and each remote cabinet controller **123** and **133**. Switches **122**, **132**, **121**, and **131** may provide redundant data links between each controller **123** and **133** and multiple motor controllers **20** (FIG. **2**).

(23) As shown in FIG. **3**, each controller **123** and **133** in channel **120** and **130** interfaces with an input/output switch **124** and **134** for sending and receiving operational data to operator controllers in RC&IS main control logic **111**. Because main control logic **111** interfaces with several other plant safety systems and operator controls, control logic **111** is typically located in or near a control room or other plant operator station. Input/output switches **124** and **134** multiplex control commands and information from both channels over main logic interfaces **140** to main control logic **111**. For example, main control logic may include two or more main RC&IS channel controllers **165** and **155**, each with two switches **161**, **162**, **151**, and **152**. Main logic interfaces **140** may be multiplexed to provide digital information and control data from both channels **120** and **130** to each switch **151**, **152**, **161**, and **162**. In this way, main RC&IS channel controllers **155** and **165** may have redundant switches that each have access to both channels associated with each control rod drive.

(24) Main RC&IS channel controllers **155** and **165** receive and implement operator feedback and input, including control element positioning commands, as well as input **200** from other plant systems, such as plant steady-state information or trip alerts. Because each main controller **155** and **165** receives redundant control and command data, and can issue redundant commands to independent channels **120** and **130**, either controller **155** and **165** may be used to report on and

operate all aspects of a control drive, regardless of failure of the other controller and/or of channel equipment or position reporting equipment.

(25) Similarly, because main RC&IS channel controllers **155** and **165** may themselves be connected and communicate through an internal bus or other plant communicative connections, controllers **155** and **165** may verify positional and operational data received as well as operator inputs between themselves. Controllers **155** and **165** may be configured to detect discrepancies between information received from either channel **120** and **130**, as well as inputs **200** or operator controls received by them both. Upon detection of a discrepancy, an operator may be notified and/or a channel may be identified as out-of-service. For example, an operator or plant control system may input a very fine control element repositioning, such as 3 mm, for a single control blade, and system **100** may receive and process the move, resulting in rod movement commands to the motor controller **20** and appropriate power and signaling transmitted to motor **19** from the motor controller **20** through motor interface **26** (FIG. **2**). Position detectors **18***a* and **18***b* (FIG. **2**) may then report digital, accurate rod repositioning information, which may be processed by the motor controller **20** and transmitted to the remote controllers **123** and **133**, which may interpret and transmit the rod repositioning information to the main controllers **165** and **155**. Main controllers **165** and **155** may verify this position data from both channels and thus detectors, and ensure it reflects the input command. Because channel controllers **155** and **165** are themselves digital, such analysis and notification or disregarding of bad channel data may be implemented as simple programming or hardware structuring in microprocessors of controllers **155** and **165**. Each channel of RC&IS controller logic, **155** and **165**, may contain one or more microprocessors, such as three redundant microprocessors R, S, and T each performing the same RC&IS logic computations independently and asynchronously of each other. Redundant microprocessors in each RC&IS control logic channel, **165** and **155** may ensure detectability and continuity of rod control logic functions in the presence of a single microprocessor failure.

(26) Example embodiment RC&IS **100** is end-to-end digital, handling data and commands received directly from digital inputs and outputs. Such digitization allows for a smaller footprint, faster analysis and action, less susceptibility to noise, vibration, heat, and radiation damage, finer motor control and monitoring, and easier handling of multiple-channel data for redundancy and verification in control rod drive operation. Further, connections among various components, such as by internal connections **115** or connections **140** between potentially remotely-situated cabinet **110** and main logic **111**, may be reliable communicative connections between systems, including internal busses on motherboards, fiber optic cables, etc.

(27) Example embodiments and methods thus being described, it will be appreciated by one skilled in the art that example embodiments may be varied and substituted through routine experimentation while still falling within the scope of the following claims. For example, although only one FMCRD **10** is shown schematically connected to example embodiment system **100**, use of multiple different FMCRDs, as well as other types of control rod drives for a variety of different reactor sizes and configurations are compatible with example embodiments and methods simply through proper dimensioning of example embodiments - and fall within the scope of the claims.

(28) Such variations are not to be regarded as departure from the scope of these claims.

Claims

1. A method of operating a fully digital control rod drive in a nuclear power plant, the method comprising: receiving a digital operation signal at a fine motion motor controller; delivering from the fine motion motor controller electrical power to a servo motor to rotate the servo motor to vertically drive a control rod according to the digital operation signal; detecting position of the control rod by a first digital position detector and a second digital position detector, wherein the first and the second digital position detectors are digital position transducers coupled to the servo

motor; and transmitting digital position information of the control rod from the first and the second digital position detectors.

2. The method of claim 1, further comprising: receiving the digital position information from the first digital position detector at a first channel of a digital rod control and information system; receiving the digital position information from the second digital position detector at a second channel of the digital rod control and information system; and verifying by the digital rod control and information system the position of the control rod from the digital position information.

3. The method of claim 2, wherein the verifying includes comparing the digital position information and the digital operational signal.

4. The method of claim 2, further comprising: at least one of notifying an operator and disabling one of the first and the second channels in response to the verifying determining an error in the digital position information.

5. The method of claim 1, wherein the first and the second digital position transducers measure to a thirty-sixth of a millimeter of the position of the control rod.

6. A method of operating a fully digital control rod drive in a nuclear power plant, the method comprising: receiving a digital operation signal at a fine motion motor controller; delivering from the fine motion motor controller electrical power to a servo motor to rotate the servo motor to vertically drive a control rod according to the digital operation signal; detecting position of the control rod by a first digital position detector and a second digital position detector; and transmitting digital position information of the control rod from the first and the second digital position detectors, wherein the fine motion motor controller does not receive signals from a digital-to-analog or an analog-to digital converter, and wherein the first and the second digital position detectors do not transmit signals to a digital-to-analog or an analog-to digital converter.

7. The method of claim 1, wherein the delivering from the fine motion motor controller electrical power to the servo motor to rotate the servo motor vertically drives the control rod in a 3 millimeter vertical increment.

8. The method of claim 1, wherein the delivering from the fine motion motor controller electrical power to the servo motor to rotate the servo motor rotates the servo motor up to a quarter of a full turn.

9. The method of claim 1 further comprising: receiving a movement command for the control rod at a digital rod control and information system; transmitting by the digital rod control and information system the digital operation signal to the control rod drive for the control rod according to the movement command; receiving at the digital rod control and information system the digital position information from the first digital position detector and the second digital position detector of the control rod drive; verifying by the digital rod control and information system a position of the control rod from the digital position information; and reporting the position of the control rod by the digital rod control and information system.

10. The method of claim 9, wherein the digital position information is digital rotation information of the servo motor.

11. The method of claim 9, wherein the verifying includes comparing the digital position information against at least one of the movement command and the digital operational signal.

12. The method of claim 9, further comprising: at least one of notifying an operator and disabling one of the first and the second channels in response to the verifying determining an error in the digital position information.

13. The method of claim 9, wherein the digital rod control and information system is in a control room remote from the control rod drive.

14. The method of claim 6, further comprising: receiving the digital position information from the first digital position detector at a first channel of a digital rod control and information system; receiving the digital position information from the second digital position detector at a second channel of the digital rod control and information system; and verifying by the digital rod control

and information system the position of the control rod from the digital position information.
15. The method of claim 14, wherein the verifying includes comparing the digital position information and the digital operational signal.
16. The method of claim 14, further comprising: at least one of notifying an operator and disabling one of the first and the second channels in response to the verifying determining an error in the digital position information.
17. The method of claim 14, wherein the first and the second digital position detectors are digital position transducers coupled to the servo motor.
18. The method of claim 17, wherein the first and the second digital position transducers measure to a thirty-sixth of a millimeter of the position of the control rod.
19. The method of claim 14, wherein the delivering from the fine motion motor controller electrical power to the servo motor to rotate the servo motor vertically drives the control rod in a **3** millimeter vertical increment.
20. The method of claim 14, wherein the delivering from the fine motion motor controller electrical power to the servo motor to rotate the servo motor rotates the servo motor up to a quarter of a full turn.